

Savannah Tourism Database System Design

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Contribution Page

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I. Introduction:

This report details a database system design, implementations, and sample queries for Savannah tourism book, with the primary goal being to take a set of semantic rules that describes a detailed list of Savannah attractions, businesses, hotels, and restaurants and convert them into a normalized universal relation database structure. The first part of this document establishes functional dependencies (FD) via twenty-six semantic rules, visualizes them in a Universal Relation Diagram, and then systematically decomposes the data through First (1NF), Second (2NF), Third (3NF), Boyce-Codd (BCNF), and Fourth (4NF) Normal Form. The second part of this document showcases the implementation of the database into a MySQL client, a database management software, including all INSERT statements and the completed relation it outputs. This document then concludes with two views of this database, a final set of seventeen queries, the results of said queries, and an appendix of meeting minutes with the database admin team.

II. Semantic Rules:

1. Each Book has an ISBN, title, publisher, edition number, year of publication, and a number of pages.
2. Each Book has several Attractions.
3. Each Book has several Businesses.
4. Each Book has several Hotels.
5. Each Book has several Restaurants.
6. Each Attraction has several Emergency Agencies.
7. Each Attraction has several Savannah area attractions.
8. Each Attraction has several Tybee Island area attractions.
9. Each Attraction has several Historic Cemeteries.
10. Each Attraction has several Historic Churches.
11. Each Attraction has several Historic Squares.
12. Each Attraction has several Public Parks.
13. Each Attraction has several Events.
14. Each Attraction has several Art attractions.
15. Each Attraction has several Sports attractions.
16. Each Attraction has several Universities.
17. Each Emergency Agency has a name and phone number.
18. Each Savannah area attraction has a name, location, phone number, website, and description.
19. Each Tybee Island area attraction has a name, location, phone number, website, and description.
20. Each Historic Cemetery has a name, location, year established, type, and description.
21. Each Historic Church has a name, phone number, location, denomination, website, and description.

22. Each Historic Square has a name, a location, a designer, a design year, architecture style, and description.
23. Each Public Park has a name, location, municipality, and description.
24. Each Calendar Event has a name, phone number, location, website, start date, and description.
25. Each Art attraction has a name, phone number, location, website, and type.
26. Each Sports attraction has several golf courses.
27. Each Sports attraction has several tennis courses.
28. Each Sports attraction has several boat ramps.
29. Each Sports attraction has several fishing piers.
30. Each Golf Course has a name, phone number, location, website, and description.
31. Each Tennis Course has a name, phone number, location, and description.
32. Each Boating Ramp has a name and location.
33. Each Fishing Pier has a name and location.
34. Each University has a name, phone number, email address, location, website, type, and description.
35. Each University has several colleges.
36. Each University has several programs.
37. Each University has several departments.
38. Each department in a university has a name and phone number.
39. Each Business has several Medical Establishments.
40. Each Business has several Pest Control Companies.
41. Each Business has several Plumbers.
42. Each Medical Establishment has a name, a specialty, a description, services offered, a location, hours of operation, website(s), and phone number(s).
43. Each Medical Establishment has several doctors.
44. Each Doctor has a name.
45. Each Doctor has several degrees.
46. Each Pest Control Company has a name, location, phone number, operating hours, specialty, payment method accepted, and description.
47. Each Plumber has a name, location, phone number, operating hours, specialty, payment method accepted, and description.
48. Each Hotel has a name, phone number, location, website, price range, booking restrictions, amenities, payment method accepted, historic area, description, and cancellation restrictions.
49. Each Restaurant has a name, phone number, location, website, food speciality, may or may not accept reservations, and description.

II.1 Conversions:

1. Each Book has an ISBN, title, publisher, edition number, year of publication, and a number of pages.
 - a. Entity: Book
 - b. Attribute(s): ISBN, title, publisher, edition, publication year, number of pages
 - c. Attribute Names (in order): ISBN, Title, Pub, Edition, PubYear, NumPgs
 - d. Entity ID: ISBN
 - e. Functional Dependency (FD): $\text{ISBN} \rightarrow \text{Title, Pub, Edition, PubYear, NumPgs}$
2. Each Book has several Attractions.
 - a. Entity: Book
 - b. Attribute(s): Attractions
 - c. Attribute Names (in order): Att# (created a unique identifier for this attribute because the attribute listed cannot serve as a unique identifier for Attractions on its own)
 - d. Entity ID: ISBN (borrowed from Semantic Rule #1)
 - e. Multivalued Dependency (MVD): $\text{ISBN} \twoheadrightarrow \text{Att\#}$
3. Each Book has several Businesses.
 - a. Entity: Book
 - b. Attribute(s): Businesses
 - c. Attribute Names (in order): BusID (created a unique identifier for this attribute because the attribute listed cannot serve as a unique identifier for Businesses on its own)
 - d. Entity ID: ISBN (borrowed from Semantic Rule #1)
 - e. Multivalued Dependency (MVD): $\text{ISBN} \twoheadrightarrow \text{BusID}$
4. Each Book has several Hotels.
 - a. Entity: Book
 - b. Attribute(s): Hotels
 - c. Attribute Names (in order): Hotel# (created a unique identifier for this attribute because the attribute listed cannot serve as a unique identifier for Hotels on its own)
 - d. Entity ID: ISBN (borrowed from Semantic Rule #1)
 - e. Multivalued Dependency (MVD): $\text{ISBN} \twoheadrightarrow \text{Hotel\#}$
5. Each Book has several Restaurants.
 - a. Entity: Book
 - b. Attribute(s): Restaurants
 - c. Attribute Names (in order): Rest# (created a unique identifier for this attribute because the attribute listed cannot serve as a unique identifier for Restaurants on its own)
 - d. Entity ID: ISBN (borrowed from Semantic Rule #1)
 - e. Multivalued Dependency (MVD): $\text{ISBN} \twoheadrightarrow \text{Rest\#}$

6. Each Attraction has several Emergency Agencies.
 - a. Entity: Attraction
 - b. Attribute(s): Emergency Agencies
 - c. Attribute Names (in order): Emergency# (created a unique identifier for this attribute because the attribute listed cannot serve as a unique identifier for Emergency Agencies on its own)
 - d. Entity ID: Att# (borrowed from Semantic Rule #2)
 - e. Multivalued Dependency (MVD): $\text{Att\#} \twoheadrightarrow \text{Emergency\#}$
7. Each Attraction has several Savannah area attractions.
 - a. Entity: Attraction
 - b. Attribute(s): Savannah Area Attractions
 - c. Attribute Names (in order): Sav_Att# (created a unique identifier for this attribute because the attribute listed cannot serve as a unique identifier for Savannah Area Attractions on its own)
 - d. Entity ID: Att# (borrowed from Semantic Rule #2)
 - e. Multivalued Dependency (MVD): $\text{Att\#} \twoheadrightarrow \text{Sav_Att\#}$
8. Each Attraction has several Tybee Island area attractions.
 - a. Entity: Attraction
 - b. Attribute(s): Tybee Island Area Attractions
 - c. Attribute Names (in order): Tyb_Att# (created a unique identifier for this attribute because the attribute listed cannot serve as a unique identifier for Tybee Island Area Attractions on its own)
 - d. Entity ID: Att# (borrowed from Semantic Rule #2)
 - e. Multivalued Dependency (MVD): $\text{Att\#} \twoheadrightarrow \text{Tyb_Att\#}$
9. Each Attraction has several Historic Cemeteries.
 - a. Entity: Attraction
 - b. Attribute(s): Historic Cemeteries
 - c. Attribute Names (in order): Cemetery# (created a unique identifier for this attribute because the attribute listed cannot serve as a unique identifier for Historic Cemeteries on its own)
 - d. Entity ID: Att# (borrowed from Semantic Rule #2)
 - e. Multivalued Dependency (MVD): $\text{Att\#} \twoheadrightarrow \text{Cemetery\#}$
10. Each Attraction has several Historic Churches.
 - a. Entity: Attraction
 - b. Attribute(s): Historic Churches
 - c. Attribute Names (in order): Church# (created a unique identifier for this attribute because the attribute listed cannot serve as a unique identifier for Historic Churches on its own)
 - d. Entity ID: Att# (borrowed from Semantic Rule #2)
 - e. Multivalued Dependency (MVD): $\text{Att\#} \twoheadrightarrow \text{Church\#}$

11. Each Attraction has several Historic Squares.
 - a. Entity: Attraction
 - b. Attribute(s): Historic Squares
 - c. Attribute Names (in order): Square# (created a unique identifier for this attribute because the attribute listed cannot serve as a unique identifier for Historic Squares on its own)
 - d. Entity ID: Att# (borrowed from Semantic Rule #2)
 - e. Multivalued Dependency (MVD): $\text{Att\#} \twoheadrightarrow \text{Square\#}$
12. Each Attraction has several Public Parks.
 - a. Entity: Attraction
 - b. Attribute(s): Parks
 - c. Attribute Names (in order): Park# (created a unique identifier for this attribute because the attribute listed cannot serve as a unique identifier for Parks on its own)
 - d. Entity ID: Att# (borrowed from Semantic Rule #2)
 - e. Multivalued Dependency (MVD): $\text{Att\#} \twoheadrightarrow \text{Park\#}$
13. Each Attraction has several Events.
 - a. Entity: Attraction
 - b. Attribute(s): Events
 - c. Attribute Names (in order): CalEvents# (created a unique identifier for this attribute because the attribute listed cannot serve as a unique identifier for Calendar of Events on its own)
 - d. Entity ID: Att# (borrowed from Semantic Rule #2)
 - e. Multivalued Dependency (MVD): $\text{Att\#} \twoheadrightarrow \text{CalEvents\#}$
14. Each Attraction has several Art attractions.
 - a. Entity: Attraction
 - b. Attribute(s): Art Attractions
 - c. Attribute Names (in order): Art# (created a unique identifier for this attribute because the attribute listed cannot serve as a unique identifier for Art Attractions on its own)
 - d. Entity ID: Att# (borrowed from Semantic Rule #2)
 - e. Multivalued Dependency (MVD): $\text{Att\#} \twoheadrightarrow \text{Art\#}$
15. Each Attraction has several Sports attractions.
 - a. Entity: Attraction
 - b. Attribute(s): Sports Attractions
 - c. Attribute Names (in order): Sports# (created a unique identifier for this attribute because the attribute listed cannot serve as a unique identifier for Sports Attractions on its own)
 - d. Entity ID: Att# (borrowed from Semantic Rule #2)
 - e. Multivalued Dependency (MVD): $\text{Att\#} \twoheadrightarrow \text{Sports\#}$

16. Each Attraction has several Universities.
 - a. Entity: Attraction
 - b. Attribute(s): Universities
 - c. Attribute Names (in order): Uni# (created a unique identifier for this attribute because the attribute listed cannot serve as a unique identifier Universities on its own)
 - d. Entity ID: Att# (borrowed from Semantic Rule #2)
 - e. Multivalued Dependency (MVD): $\text{Att\#} \twoheadrightarrow \text{Uni\#}$
17. Each Emergency Agency has a name and phone number.
 - a. Entity: Emergency Agency
 - b. Attribute(s): Name, Phone number
 - c. Attribute Names (in order): Emergency_Name, Emergency_Phone
 - d. Entity ID: Emergency# (borrowed from Semantic Rule #6)
 - e. Functional Dependency (FD): $\text{Emergency\#} \rightarrow \text{Emergency_Name, Emergency_Phone}$
18. Each Savannah area attraction has a name, location, phone number, website, and description.
 - a. Entity: Savannah Area Attraction
 - b. Attribute(s): Name, Location, Phone number, Website, Description
 - c. Attribute Names (in order): Sav_Att_Name, Sav_Att_Phone, Sav_Att_Loc, Sav_Att_Web, Sav_Att_Descript
 - d. Entity ID: Sav_Att# (borrowed from Semantic Rule #7)
 - e. Functional Dependency (FD): $\text{Sav_Att\#} \rightarrow \text{Sav_Att_Name, Sav_Att_Phone, Sav_Att_Loc, Sav_Att_Web, Sav_Att_Descript}$
19. Each Tybee Island area attraction has a name, location, phone number, website, and description.
 - a. Entity: Tybee Island Area Attraction
 - b. Attribute(s): Name, Location, Phone number, Website, Description
 - c. Attribute Names (in order): Tyb_Att_Name, Tyb_Att_Loc, Tyb_Att_Phone, Tyb_Att_Web, Tyb_Att_Descript
 - d. Entity ID: Tyb_Att# (borrowed from Semantic Rule #8)
 - e. Functional Dependency (FD): $\text{Tyb_Att\#} \rightarrow \text{Tyb_Att_Name, Tyb_Att_Loc, Tyb_Att_Phone, Tyb_Att_Web, Tyb_Att_Descript}$
20. Each Historic Cemetery has a name, location, year established, type, and description.
 - a. Entity: Historic Cemetery
 - b. Attribute(s): Name, Location, Year Established, Type, Description
 - c. Attribute Names (in order): Cemetery_Name, Cemetery_Loc, Cemetery_EstYear, Cemetery_Type, Cemetery_Descript
 - d. Entity ID: Cemetery# (borrowed from Semantic Rule #9)

- e. Functional Dependency (FD): Cemetery# $\rightarrow$ Cemetery_Name, Cemetery_Loc, Cemetery_EstYear, Cemetery_Type, Cemetery_Descript
21. Each Historic Church has a name, phone number, location, denomination, website, and description.
- a. Entity: Historic Church
 - b. Attribute(s): Name, phone number, location, denomination, website, description
 - c. Attribute Names (in order): Church_Name, Church_Phone, Church_Loc, Church_Denom, Church_Web, Church_Descript
 - d. Entity ID: Church# (borrowed from Semantic Rule #10)
 - e. Functional Dependency (FD): Church# $\rightarrow$ Church_Name, Church_Phone, Church_Loc, Church_Denom, Church_Web, Church_Descript
22. Each Historic Square has a name, location, inspiration, design year, and description.
- a. Entity: Historic Square
 - b. Attribute(s): Name, Location, Inspiration, Design year, Description
 - c. Attribute Names (in order): Square_Name, Square_Loc, Square_Inspiration, Square_DesignYear, Square_Descript
 - d. Entity ID: Square# (borrowed from Semantic Rule #11)
 - e. Functional Dependency (FD): Square# $\rightarrow$ Square_Name, Square_Loc, Square_Inspiration, Square_DesignYear, Square_Descript
23. Each Public Park has a name, location, municipality, and description.
- a. Entity: Park
 - b. Attribute(s): Name, Location, Municipality, Description
 - c. Attribute Names (in order): Park_Name, Park_Loc, Park_Munic, Park_Descript
 - d. Entity ID: Park# (borrowed from Semantic Rule #12)
 - e. Functional Dependency (FD): Park# $\rightarrow$ Park_Name, Park_Loc, Park_Munic, Park_Descript
24. Each Calendar Event has a name, phone number, location, website, start date, and description.
- a. Entity: Calendar Event
 - b. Attribute(s): Name, Phone number, Location, Website, Start Date, and Description
 - c. Attribute Names (in order): CalEvent_Name, CalEvent_Phone, CalEvent_Loc, CalEvent_Web, CalEvent_StartDate, CalEvent_Descript
 - d. Entity ID: CalEvent# (borrowed from Semantic Rule #13)
 - e. Functional Dependency (FD): CalEvent# $\rightarrow$ CalEvent_Name, CalEvent_Phone, CalEvent_Loc, CalEvent_Web, CalEvent_StartDate, CalEvent_Descript
25. Each Art attraction has a name, phone number, location, website, and type.
- a. Entity: Art Attraction
 - b. Attribute(s): Name, Phone number, Location, Website, Type
 - c. Attribute Names (in order): Art_Name, Art_Phone, Art_Loc, Art_Web, Art_Type
 - d. Entity ID: Art# (borrowed from Semantic Rule #14)

- e. Functional Dependency (FD): $\text{Art\#} \rightarrow \text{Art_Name}, \text{Art_Phone}, \text{Art_Loc}, \text{Art_Web}, \text{Art_Type}$
26. Each Sports attraction has several golf courses.
- a. Entity: Sports Attraction
 - b. Attribute(s): Golf Courses
 - c. Attribute Names (in order): Golf# (created a unique identifier for this attribute because the attribute listed cannot serve as a unique identifier for Golf Courses on its own)
 - d. Entity ID: Sports# (borrowed from Semantic Rule #15)
 - e. Multivalued Dependency (MVD): $\text{Sports\#} \twoheadrightarrow \text{Golf\#}$
27. Each Sports attraction has several tennis courts.
- a. Entity: Sports Attraction
 - b. Attribute(s): Tennis Courts
 - c. Attribute Names (in order): Tennis# (created a unique identifier for this attribute because the attribute listed cannot serve as a unique identifier for Tennis Courts on its own)
 - d. Entity ID: Sports# (borrowed from Semantic Rule #15)
 - e. Multivalued Dependency (MVD): $\text{Sports\#} \twoheadrightarrow \text{Tennis\#}$
28. Each Sports attraction has several boat ramps.
- a. Entity: Sports Attraction
 - b. Attribute(s): Boat Ramps
 - c. Attribute Names (in order): Ramp# (created a unique identifier for this attribute because the attribute listed cannot serve as a unique identifier for Boat Ramps on its own)
 - d. Entity ID: Sports# (borrowed from Semantic Rule #15)
 - e. Multivalued Dependency (MVD): $\text{Sports\#} \twoheadrightarrow \text{Ramp\#}$
29. Each Sports attraction has several fishing piers.
- a. Entity: Sports attraction
 - b. Attribute(s): Fishing Piers
 - c. Attribute Names (in order): Pier# (created a unique identifier for this attribute because the attribute listed cannot serve as a unique identifier for Fishing Piers on its own)
 - d. Entity ID: Sports# (borrowed from Semantic Rule #15)
 - e. Multivalued Dependency (MVD): $\text{Sports\#} \twoheadrightarrow \text{Pier\#}$
30. Each Golf Course has a name, phone number, location, website, and description.
- a. Entity: Golf Course
 - b. Attribute(s): Name, Phone number, Location, Website, Description
 - c. Attribute Names (in order): Golf_Name, Golf_Phone, Golf_Loc, Golf_Web, Golf_Descript
 - d. Entity ID: Golf# (borrowed from Semantic Rule #26)

- e. Functional Dependency (FD): $\text{Golf\#} \rightarrow \text{Golf_Name}, \text{Golf_Phone}, \text{Golf_Loc}, \text{Golf_Web}, \text{Golf_Descript}$
- 31. Each Tennis Court has a name, phone number, location, and description.
 - a. Entity: Tennis court
 - b. Attribute(s): Name, Phone number, Location, Description
 - c. Attribute Names (in order): Tennis_Name, Tennis_Phone, Tennis_Loc, Tennis_Descript
 - d. Entity ID: Tennis# (borrowed from Semantic Rule #27)
 - e. Functional Dependency (FD): $\text{Tennis\#} \rightarrow \text{Tennis_Name}, \text{Tennis_Phone}, \text{Tennis_Loc}, \text{Tennis_Descript}$
- 32. Each Boat Ramp has a name and location.
 - a. Entity: Boat Ramp
 - b. Attribute(s): Name, Location
 - c. Attribute Names (in order): Ramp_Name, Ramp_Loc
 - d. Entity ID: Ramp# (borrowed from Semantic Rule #28)
 - e. Functional Dependency (FD): $\text{Ramp\#} \rightarrow \text{Ramp_Name}, \text{Ramp_Loc}$
- 33. Each Fishing Pier has a name and location.
 - a. Entity: Fishing Pier
 - b. Attribute(s): Name, location
 - c. Attribute Names (in order): Pier_Name, Pier_Loc
 - d. Entity ID: Pier# (borrowed from Semantic Rule #29)
 - e. Functional Dependency (FD): $\text{Pier\#} \rightarrow \text{Pier_Name}, \text{Pier_Loc}$
- 34. Each University has a name, phone number, email address, location, website, type, and description.
 - a. Entity: University
 - b. Attribute(s): Name, Phone number, Email address, Location, Website, Type, Description
 - c. Attribute Names (in order): Uni_Name, Uni_Phone, Uni_Email, Uni_Loc, Uni_Web, Uni_Type, Uni_Descript
 - d. Entity ID: Uni# (borrowed from Semantic Rule #16)
 - e. Functional Dependency (FD): $\text{Uni\#} \rightarrow \text{Uni_Name}, \text{Uni_Phone}, \text{Uni_Email}, \text{Uni_Loc}, \text{Uni_Web}, \text{Uni_Type}, \text{Uni_Descript}$
- 35. Each University has several colleges.
 - a. Entity: University
 - b. Attribute(s): Colleges
 - c. Attribute Names (in order): Uni_College
 - d. Entity ID: Uni# (borrowed from Semantic Rule #16)
 - e. Multivalued Dependency (MVD): $\text{Uni\#} \twoheadrightarrow \text{Uni_College}$
- 36. Each University has several programs.
 - a. Entity: University

- b. Attribute(s): Programs
 - c. Attribute Names (in order): Uni_Program
 - d. Entity ID: Uni# (borrowed from Semantic Rule #16)
 - e. Multivalued Dependency (MVD): $\text{Uni}\# \twoheadrightarrow \text{Uni_Program}$
37. Each University has several departments.
- a. Entity: University
 - b. Attribute(s): Departments
 - c. Attribute Names (in order): UniDept# (created a unique identifier for this attribute because the attribute listed cannot serve as a unique identifier for Departments on its own)
 - d. Entity ID: Uni# (borrowed from Semantic Rule #16)
 - e. Multivalued Dependency (MVD): $\text{Uni}\# \twoheadrightarrow \text{UniDept}\#$
38. Each department in a university has a name and phone number.
- a. Entity: Department
 - b. Attribute(s): Name, Phone number
 - c. Attribute Names (in order): UniDept_Name, UniDept_Phone
 - d. Entity ID: Uni_Dept# (borrowed from Semantic Rule #37)
 - e. Functional Dependency (FD): $\text{Uni_Dept}\# \rightarrow \text{UniDept_Name}, \text{UniDept_Phone}$
39. Each Business has several Medical Establishments.
- a. Entity: Business
 - b. Attribute(s): Medical Establishments
 - c. Attribute Names (in order): Med# (created because the current attribute cannot serve as a unique identifier for Medical Establishments on its own)
 - d. Entity ID: BusID (borrowed from Semantic Rule #3)
 - e. Multivalued Dependency (MVD): $\text{BusID} \twoheadrightarrow \text{Med}\#$
40. Each Business has several Pest Control Companies.
- a. Entity: Business
 - b. Attribute(s): Pest Control Companies
 - c. Attribute Names (in order): Pest# (created because the current attribute cannot serve as a unique identifier for Pest Control Companies on its own)
 - d. Entity ID: BusID (borrowed from Semantic Rule #3)
 - e. Multivalued Dependency (MVD): $\text{BusID} \twoheadrightarrow \text{Pest}\#$
41. Each Business has several Plumbers.
- a. Entity: Business
 - b. Attribute(s): Plumbers
 - c. Attribute Names (in order): Plumb# (created because the current attribute cannot serve as a unique identifier for Plumbers on its own)
 - d. Entity ID: BusID (borrowed from Semantic Rule #3)
 - e. Multivalued Dependency (MVD): $\text{BusID} \twoheadrightarrow \text{Plumb}\#$

42. Each Medical Establishment has a name, specialty, description, services offered, location, hours of operation, website, and phone number.
- Entity: Medical Establishment
 - Attribute(s): Name, Specialty, Description, Services, Location, Operational hours, Website, Phone number
 - Attribute Names (in order): Med_Name, Med_Specialty, Med_Descript, Med_Services, Med_Loc, Med_OpHours, Med_Web, Med_Phone
 - Entity ID: Med# (borrowed from Semantic Rule #39)
 - Functional Dependency (FD): Med# $\rightarrow$ Med_Name, Med_Specialty, Med_Descript, Med_Services, Med_Loc, Med_OpHours, Med_Web, Med_Phone
43. Each Medical Establishment has several doctors.
- Entity: Medical Establishment
 - Attribute(s): Doctors
 - Attribute Names (in order): Doc# (created because the current attribute cannot serve as a unique identifier for Doctors on its own)
 - Entity ID: Med# (borrowed from Semantic Rule #39)
 - Multivalued Dependency (MVD): Med# $\twoheadrightarrow$ Doc#
44. Each Doctor has a name.
- Entity: Doctor
 - Attribute(s): Name
 - Attribute Names (in order): Doc_Name
 - Entity ID: Doc# (borrowed from Semantic Rule #43)
 - Functional Dependency (FD): Doc# $\rightarrow$ Doc_Name#
45. Each Doctor has several degrees.
- Entity: Doctor
 - Attribute(s): Degrees
 - Attribute Names (in order): Doc_Degree
 - Entity ID: Doc# (borrowed from Semantic Rule #43)
 - Multivalued Dependency (MVD): Doc# $\twoheadrightarrow$ Doc_Degree
46. Each Pest Control Company has a name, location, phone number, operating hours, specialty, payment method accepted, and description.
- Entity: Pest Control Company
 - Attribute(s): Name, Location, Phone number, Operating hours, Specialty, Accepted payment method, and Description
 - Attribute Names (in order): Pest_Name, Pest_Loc, Pest_Phone, Pest_OpHours, Pest_Specialty, Pest_AccPmt, Pest_Descript
 - Entity ID: Pest# (borrowed from Semantic Rule #40)
 - Functional Dependency (FD): Pest# $\rightarrow$ Pest_Name, Pest_Loc, Pest_Phone, Pest_OpHours, Pest_Specialty, Pest_AccPmt, Pest_Descript

47. Each Plumber has a name, location, phone number, operating hours, specialty, payment method accepted, and description.
- Entity: Plumber
 - Attribute(s): Name, Location, Phone number, Operating hours, Specialty, Accepted payment method, Description
 - Attribute Names (in order): Plumb_Name, Plumb_Loc, Plumb_Phone, Plumb_OpHours, Plumb_Specialty, Plumb_AccPmt, Plumb_Descript
 - Entity ID: Plumb# (borrowed from Semantic Rule #41)
 - Functional Dependency (FD): $\text{Plumb\#} \rightarrow \text{Plumb_Name}, \text{Plumb_Loc}, \text{Plumb_Phone}, \text{Plumb_OpHours}, \text{Plumb_Specialty}, \text{Plumb_AccPmt}, \text{Plumb_Descript}$
48. Each Hotel has a name, phone number, location, website, amenities, price range, booking restrictions, cancellation restrictions, amenities, payment method accepted, historic area, and description.
- Entity: Hotel
 - Attribute(s): Name, Phone number, Location, Website, Amenities, Price range, Booking restriction, Cancellation restrictions, Amenities, Accepted payment method, Historic area, Description
 - Attribute Names (in order): Hotel_Name, Hotel_Phone, Hotel_Loc, Hotel_Web, Hotel_PriceRange, Hotel_BookRestric, Hotel_CancelRestric, Hotel_Amenities, Hotel_AccPmt, Hotel_HisArea, Hotel_Descript
 - Entity ID: Hotel# (borrowed from Semantic Rule #4)
 - Functional Dependency (FD): $\text{Hotel\#} \rightarrow \text{Hotel_Name}, \text{Hotel_Phone}, \text{Hotel_Loc}, \text{Hotel_Web}, \text{Hotel_Ameni}, \text{Hotel_PriceRange}, \text{Hotel_BookRestric}, \text{Hotel_CancelRestric}, \text{Hotel_Amenities}, \text{Hotel_AccPmt}, \text{Hotel_HisArea}, \text{Hotel_Descript}$
49. Each Restaurant has a name, phone number, location, website, operating hours, cuisine type, may or may not accept reservations, and description.
- Entity: Restaurant
 - Attribute(s): Name, Phone number, Location, Website, Operating hours, Cuisine type, Reservations, Description
 - Attribute Names (in order): Rest_Name, Rest_Phone, Rest_Loc, Rest_Web, Rest_OpHours, Cuisine_Type, Rest_Reserve, Rest_Descript
 - Entity ID: Rest# (borrowed from Semantic Rule #5)
 - Functional Dependency (FD): $\text{Rest\#} \rightarrow \text{Rest_Name}, \text{Rest_Phone}, \text{Rest_Loc}, \text{Rest_Web}, \text{Rest_OpHours}, \text{Cuisine_Type}, \text{Rest_Reserve}, \text{Rest_Descript}$

II.2 List of Attributes:

Attribute Name	Attribute Description	Example of Attribute
Art_Loc	Art Location	204 W St. Julian St.
Art_Name	Art Name	City Market Art Center
Art_Phone	Art Phone Number	912-232-4903
Art_Type	Art Type (Concert, Tea, etc)	Concert
Art_Web	Art Website	www.savannahcitymarket.com
Art#	Art Identification	ART001
Att#	Attraction Identification	A001
BusID	Business Identification	BUS001
CalEvent_Descript	Event Description	Savannah Philharmonic Chorus, Sara Peebles, Ian O'Brien
CalEvent_Loc	Event Location	Trustees Theater
CalEvent_Name	Calendar of Events Event Name	A Night in Vienna
CalEvent_Phone	Event Phone	912-525-5050
CalEvent_StartDate	Event Start Date	2/1/2013
CalEvent_Web	Event Website	www.savannahboxoffice.com
CalEvent#	Calendar of Events Identification	CA001

Attribute Name	Attribute Description	Example of Attribute
Cemetery_Descript	Cemetery Description	Conrad Alken, Johnny Mercer, oaks, statuary, headstones
Cemetery_EstYear	Cemetery Year Established	1750
Cemetery_Loc	Cemetery Location	Wheaton St to Bonaventure Rd.
Cemetery_Name	Cemetery Name	Bonaventure Cemetery
Cemetery_Type	Cemetery Type	Historic
Cemetery#	Cemetery Identification	HCE01
Church_Denom	Church Denomination	Baptist
Church_Descript	Church Description	Organized in 1800, chartered in 1801 by Gov. Josiah Tattnall Jr, moved in 1833
Church_Loc	Church Location	233 Bull St.
Church_Name	Church Name	First Baptist Church
Church_Phone	Church Phone	912-234-2671
Church_Web	Church Website	www.christchurchsavannah.org
Church#	Church Identification	HCH01
Cuisine_Type	Restaurant Cuisine Offered	Fast Food, Seafood
Doc_Degree	Doctor's Degree(s)	M.D.

Attribute Name	Attribute Description	Example of Attribute
Doc_Name	Doctor's Name	Robert W. Beard
Doc#	Doctor Identification Number	DR001
Edition	Book Edition Number	2nd
Emergency_Name	Emergency Name	Chatham Emergency Management Agency
Emergency_Phone	Emergency Phone Number	912-201-4500
Emergency#	Emergency Identification	E1000
Golf_Descript	Golf Course Description	27 holes, par 72
Golf_Loc	Golf Course Location	One Shorty Cooper Dr.
Golf_Name	Golf Course Name	Bacon Park Golf Course
Golf_Phone	Golf Course Phone Number	912-354-2625
Golf_Web	Golf Course Website	www.baconparkgolf.com
Golf#	Golf Course Identification	GC001
Hotel_AccPmt	Hotel Payment Accepted	Visa, Mastercard, American Express
Hotel_Amenities	Hotel Amenities Available	Heated outdoor pool, whirlpool, exercise room
Hotel_BookRestrict	Restrictions on Booking Hotel	2-night minimum on Fri-Sat; 72-hour cancellation or first night charged

Attribute Name	Attribute Description	Example of Attribute
Hotel_CancelRestric	Hotel Restrictions on Cancellation	Cancellation fee imposed
Hotel_Descript	Hotel Description	Boutique B&B with antique décor
Hotel_HisArea	Hotel Historic Area	Savannah
Hotel_Loc	Hotel Location	14 E Oglethorpe Ave, Savannah, GA 31401
Hotel_Name	Hotel Name	Ballastone Inn
Hotel_Phone	Hotel Phone Number	(912) 555-0142
Hotel_PriceRange	Hotel Price Range	\$\$\$ (200–350 per night)
Hotel_Web	Hotel Website	www.hotelname.com
Hotel#	Hotel Identification	H0001
ISBN	International Serial Book Number	1234-53-456-78
Med_Descript	Medical Establishment Description	Comprehensive Ophthalmology & Emergency Services
Med_Loc	Medical Establishment Location	5354 Reynold's St., Suite 225, Candler Professional Building, Savannah, GA 31405
Med_Name	Medical Establishment Name	The Neurological Institute of Savannah & Center for Spine
Med_OpHours	Medical Establish Operation Hours	9:00-5:00 Mon, Tues, Thurs, Fri; 9:00-1:00 Wed

Attribute Name	Attribute Description	Example of Attribute
Med_Phone	Medical Establishment Phone Number	912-352-3120
Med_Services	Medical Establishment Services	Cataract Surgery, LASIK & PRK, Multifocal Lens Surgery
Med_Specialty	Medical Specialty	Ophthalmology (Eye)
Med_Web	Medical Establishment Website	www.myvisualbest.co m
Med#	Medical Establishment Identification	12
NumPgs	Total number of pages in a book	72
Park_Descript	Park Description	1.2 acres, playground, athletic field, picnic area, spray pool
Park_Loc	Park Location	2100 Texas/Ohio Ave
Park_Munic	Park Municipality	City
Park_Name	Park Name	Avondale Park
Park#	Park Identification	P1000
Pest_AccPmt	Pest Control Payment Accepted	Discover
Pest_Descript	Pest Control Description	Certified and Insured for Termite Inspections, Lower Rates, Superior Service
Pest_Loc	Pest Control Location	Savannah

Attribute Name	Attribute Description	Example of Attribute
Pest_Name	Pest Control Name	Cox Termite and Pest Control
Pest_OpHours	Pest Control Hours of Operation	M-Su 24hr
Pest_Phone	Pest Control Phone Number	912-651-9000
Pest_Specialty	Pest Control Specialty	Roaches, Ants, Termites
Pest#	Pest Control Identification	PC005
Pier_Loc	Fishing Pier Location	Island Expy
Pier_Name	Fishing Pier Name	Frank W. Spencer
Pier#	Fishing Pier Identification	FP001
Plumb_AccPmt	Plumber Company Payment Accepted	Mastercard
Plumb_Descript	Plumber Company Description	Guaranteed workmanship, on-time and rapid response
Plumb_Loc	Plumber Company Location	5019 Ogeechee Rd., Savannah, GA 31312
Plumb_Name	Plumber Name	Thompson & Thompson Service Group
Plumb_OpHours	Plumber Company Hours of Operation	M-Su 24hr
Plumb_Phone	Plumber Company Phone	912-373-8511
Plumb_Specialty	Plumber Company Specialty	Water Heaters, Sump Pumps
Plumb#	Plumber Company Identification	PL002

Attribute Name	Attribute Description	Example of Attribute
Pub	Publisher of book	Focus
PubYear	Year of Publication	2020
Ramp_Loc	Boat Ramp Location	6851 Ogeechee Rd
Ramp_Name	Boat Ramp Name	Kings Ferry
Ramp#	Boat Ramp Identification	BR010
Rest_Descript	Restaurant Description	Casual waterfront spot known for shrimp and grits
Rest_Loc	Restaurant Location	412 W River St, Savannah, GA 31401
Rest_Name	Restaurant Name	River Street Seafood Co.
Rest_OpHours	Restaurant Hours of Operation	Mon–Thu 11:00–21:00; Fri–Sat 11:00–23:00; Sun 10:00–15:00 (brunch)
Rest_Phone	Restaurant Name	(912) 555-0198
Rest_Reserve	Restaurant Accepts Reservations?	Yes, via OpenTable; walk-ins accepted
Rest_Web	Restaurant Website	www.restaurantname.com
Rest#	Restaurant Identification	REST0001
Sav_Att_Descript	Savannah Attraction Description	11 ft bronze monument of a black family embracing in shackles
Sav_Att_Loc	Savannah Attraction Location	Rousakis Plaza/River St.

Attribute Name	Attribute Description	Example of Attribute
Sav_Att_Name	Savannah Attraction Name	African-American Monument
Sav_Att_Phone	Savannah Attraction Phone Number	912-233-6854
Sav_Att_Web	Savannah Attraction Website	www.andrewlowhouse.com
Sav_Att#	Savannah Attraction Identification	SA100
Sports#	Sports Identification	S001
Square_Descript	Square Description	Only a small portion remains
Square_DesignYear	Square Designed Year	1851
Square_Inspiration	Square Inspiration	John C. Calhoun
Square_Loc	Square Location	Abercorn St between Taylor and Gordon Sts
Square_Name	Square Name	Calhoun Square
Square#	Square Identification	HS100
Tennis_Descript	Tennis Court Description	16 courts, all lit
Tennis_Loc	Tennis Court Location	6262 Skidaway Rd.
Tennis_Name	Tennis Court Name	Bacon Parl
Tennis_Phone	Tennis Court Phone	912-351-3850
Tennis#	Tennis Court Identification	TC001

Attribute Name	Attribute Description	Example of Attribute
Title	Title of book	Glorious Savannah
Tyb_Att_Descript	Tybee Island Attraction Description	One of last coastal artillery batteries built along East Coast
Tyb_Att_Loc	Tybee Island Attraction Location	US Highway 80
Tyb_Att_Name	Tybee Island Attraction Name	Tybee Island Historical Society - Fort Screven
Tyb_Att_Phone	Tybee Island Attraction Phone	912-786-5801
Tyb_Att_Web	Tybee Island Attraction Website	www.tybeelighthouse.org
Tyb_Att#	Tybee Island Attraction Identification	TB100
Uni_College	University Colleges	Business Administration
Uni_Dept#	University Business Departments	UD0201
Uni_Descript	University Description	Catholic, liberal arts, chartered in 1889
Uni_Email	University Email Address	savannah@saintleo.edu
Uni_Loc	University Location	1332 Southern Dr
Uni_Name	University Name	Georgia Southern
Uni_Phone	University Phone Number	912-478-4636
Uni_Program	University Programs	Science and Technology
Uni_Type	University Type (2/4 year, etc)	2 year

Attribute Name	Attribute Description	Example of Attribute
Uni_Web	University Website	www.georgiasouthern.edu
Uni#	University Identification	UN002
UniDept_Name	University Business Department Names	Admissions
UniDept_Phone	University Business Department Phone Numbers	912-478-5391

III. List of Functional Dependencies (FD):

- ISBN → Title, Pub, Edition, PubYear, NumPgs
- Emergency# → Emergency_Name, Emergency_Phone
- Sav_Att# → Sav_Att_Name, Sav_Att_Phone, Sav_Att_Loc, Sav_Att_Web, Sav_Att_Descript
- Tyb_Att# → Tyb_Att_Name, Tyb_Att_Loc, Tyb_Att_Phone, Tyb_Att_Web, Tyb_Att_Descript
- Cemetery# → Cemetery_Name, Cemetery_Loc, Cemetery_EstYear, Cemetery_Type, Cemetery_Descript
- Church# → Church_Name, Church_Phone, Church_Loc, Church_Denom, Church_Web, Church_Descript
- Square# → Square_Name, Square_Loc, Square_Inspiration, Square_DesignYear, Square_Descript
- Park# → Park_Name, Park_Loc, Park_Munic, Park_Descript
- CalEvent# → CalEvent_Name, CalEvent_Phone, CalEvent_Loc, CalEvent_Web, CalEvent_StartDate, CalEvent_Descript
- Art# → Art_Name, Art_Phone, Art_Loc, Art_Web, Art_Type
- Golf# → Golf_Name, Golf_Phone, Golf_Loc, Golf_Web, Golf_Descript
- Tennis# → Tennis_Name, Tennis_Phone, Tennis_Loc, Tennis_Descript
- Ramp# → Ramp_Name, Ramp_Loc
- Pier# → Pier_Name, Pier_Loc
- Uni# → Uni_Name, Uni_Phone, Uni_Email, Uni_Loc, Uni_Web, Uni_Type, Uni_Descript
- Uni_Dept# → UniDept_Name, UniDept_Phone
- Med# → Med_Name, Med_Specialty, Med_Descript, Med_Services, Med_Loc, Med_OpHours, Med_Web, Med_Phone
- Doc# → Doc_Name

- Pest# → Pest_Name, Pest_Loc, Pest_Phone, Pest_OpHours, Pest_Specialty, Pest_AccPmt, Pest_Descript
- Plumb# → Plumb_Name, Plumb_Loc, Plumb_Phone, Plumb_OpHours, Plumb_Specialty, Plumb_AccPmt, Plumb_Descript
- Hotel# → Hotel_Name, Hotel_Phone, Hotel_Loc, Hotel_Web, Hotel_PriceRange, Hotel_BookRestric, Hotel_CancelRestric, Hotel_Amenities, Hotel_AccPmt, Hotel_HisArea, Hotel_Descript
- Rest# → Rest_Name, Rest_Phone, Rest_Loc, Rest_Web, Rest_OpHours, Cuisine_Type, Rest_Reserve, Rest_Descript

IV. List of Multivalued Dependencies (MVD):

- ISBN →→ Att#
- ISBN →→ BusID
- ISBN →→ Hotel#
- ISBN →→ Rest#
- Att# →→ Emergency#
- Att# →→ Sav_Att#
- Att# →→ Tyb_Att#
- Att# →→ Cemetery#
- Att# →→ Church#
- Att# →→ Square#
- Att# →→ Park#
- Att# →→ CalEvents#
- Att# →→ Art#
- Att# →→ Sports#
- Att# →→ Uni#
- Sports# →→ Golf#
- Sports# →→ Tennis#
- Sports# →→ Ramp#
- Sports# →→ Pier#
- Uni# →→ Uni_College
- Uni# →→ Uni_Program
- Uni# →→ UniDept#
- BusID →→ Med#
- BusID →→ Pest#
- BusID →→ Plumb#
- Med# →→ Doc#
- Doc# →→ Doc_Degree

V. Universal Relation Diagram:

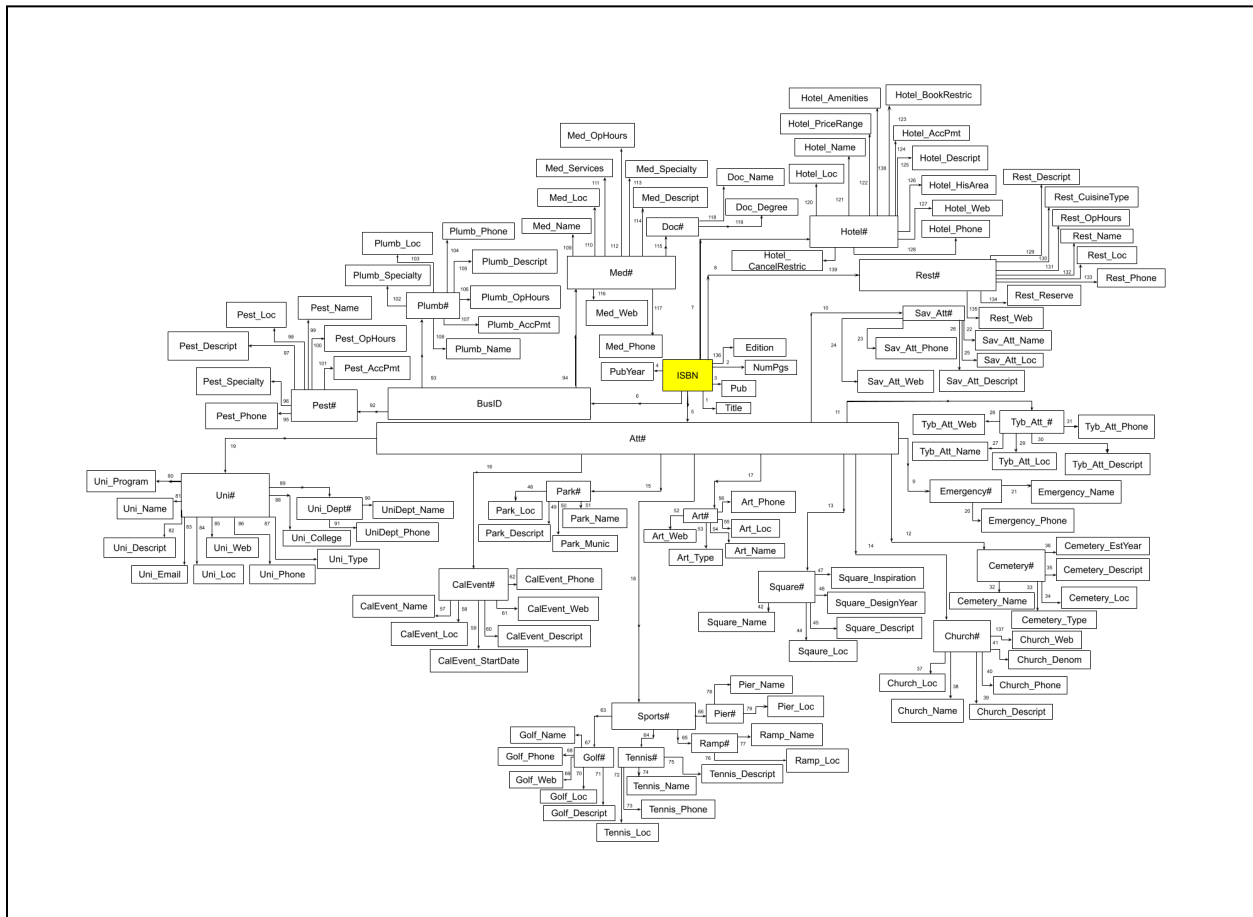


Figure 1: Initial FD Diagram

IV.1 Key for the Relation:

The **ISBN** attribute will act as the primary key for this relation because (a) it meets both the uniqueness condition and the minimality condition and (b) it can reach all the other (non-prime) attributes in the diagram.

IV.2 Reduced FD Diagram:

The following diagram is a reduced version of the initial FD diagram based off of the following redundancy rules:

1. Primary key determines every non-prime attribute (by default), so arrows initiating from the primary key are removed *unless* their removal orphans a non-prime attribute from the key.
2. All arrows representing the default transitive dependencies are removed.

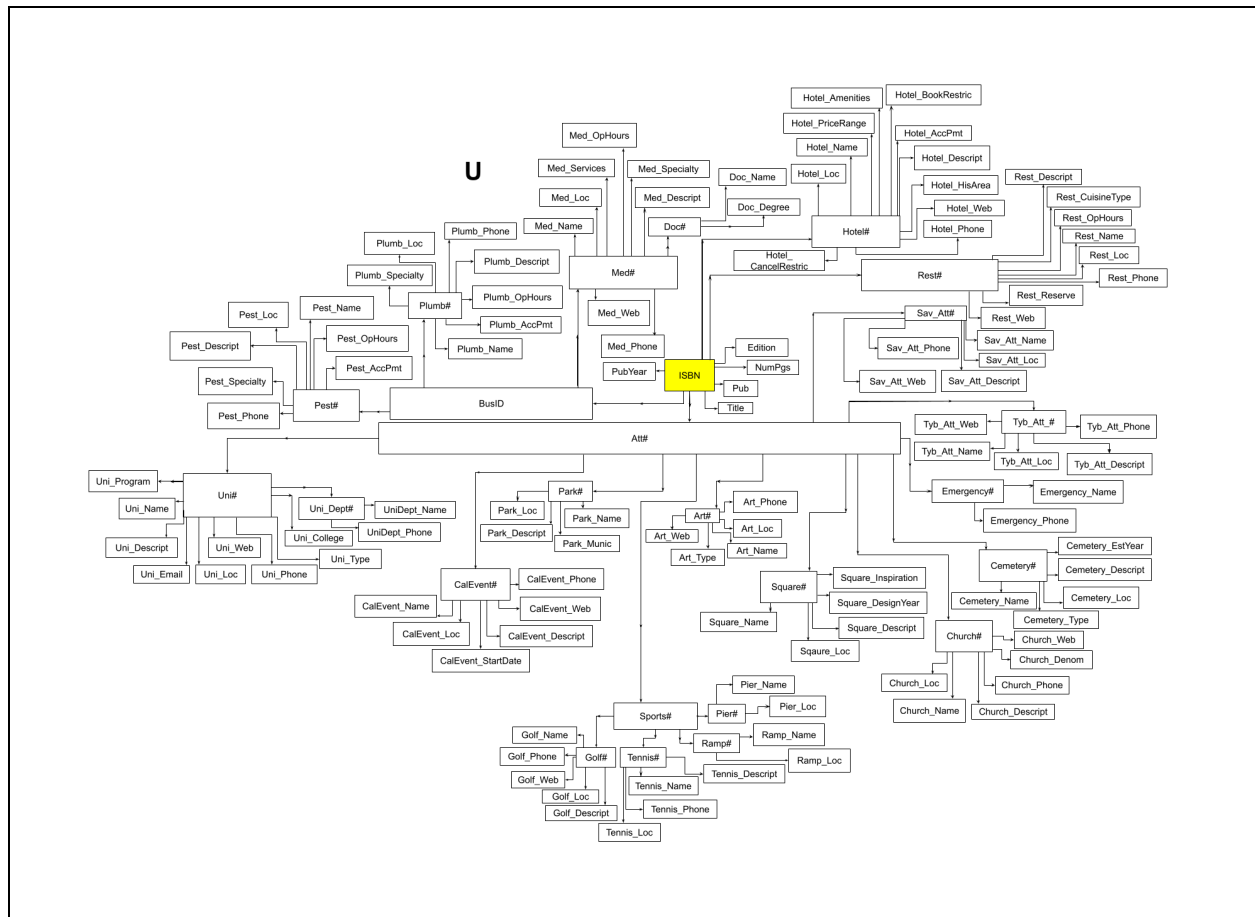


Figure 3: Diagram of the relation in 1NF (same as Figure 1 & Figure 2)

Reasonings: No additional changes were made to the diagram. Since all attributes are atomic, the Relation U is already in 1NF.

Second Normal Form (2NF):

Definition: A relation is in Second Normal Form, if (a) it is in 1NF and (b) every non-prime attribute is fully functionally dependent on the primary key.

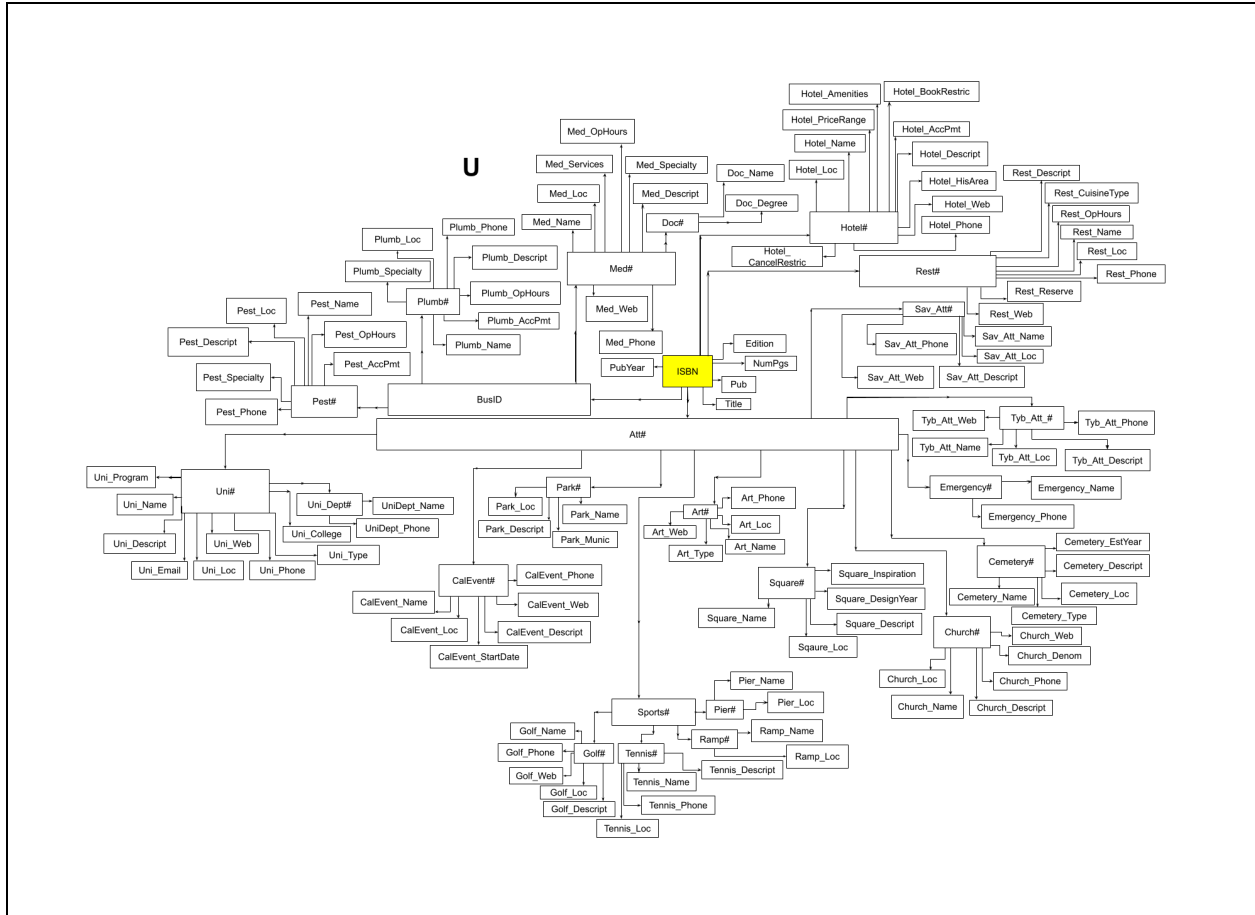


Figure 4: Diagram of the relation in 2NF (Same as Figures 1, 2, & 3)

Reasonings: No additional changes were made to the diagram. **The non-prime attributes in Relation U are fully functionally dependent on the entire primary key (ISBN).**

Third Normal Form (3NF):

Definition: A relation is in Third Normal Form, if (a) it is in 2NF and (b) every non-prime attribute is nontransitively dependent on the primary key.

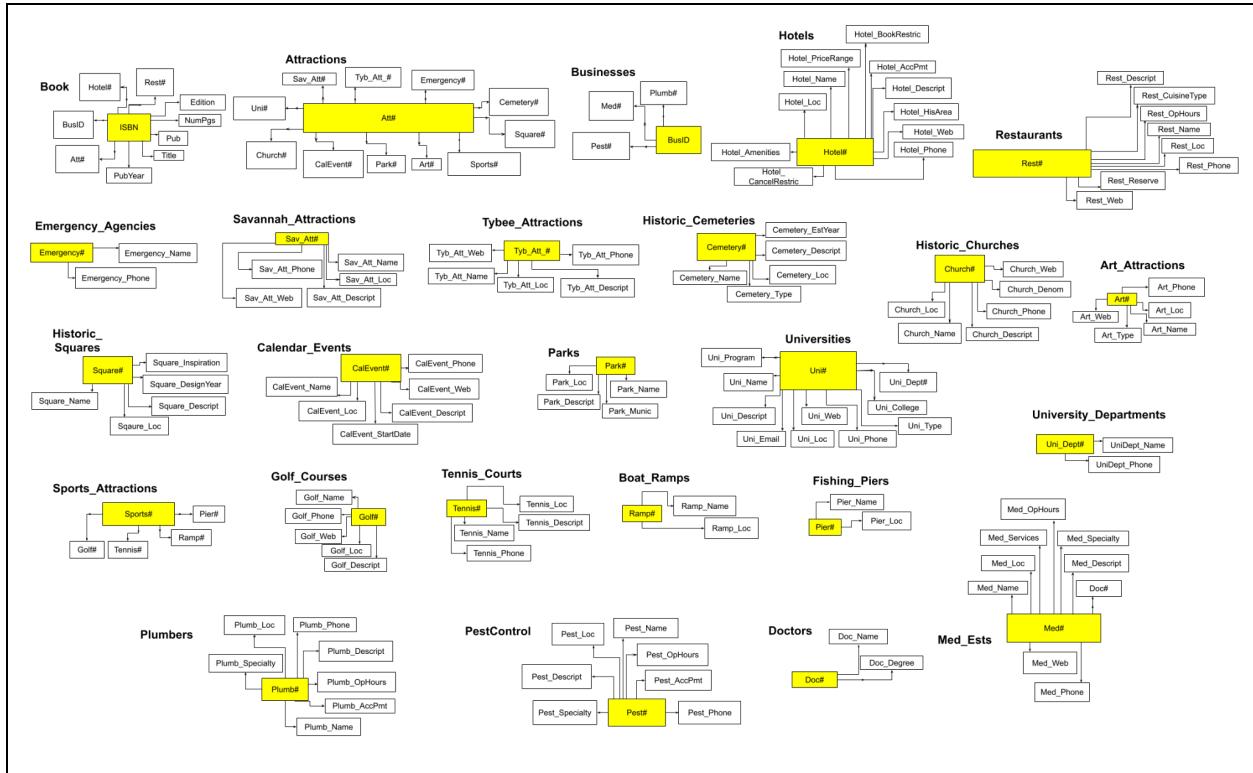


Figure 6: Diagram of the relations in BCNF (same as Figure 5)

Reasonings: No additional changes were made to the diagram as **every determinant is a candidate key of each relation.**

Fourth Normal Form (4NF):

Definition: A relation is in Fourth Normal Form if (a) it is in BCNF and (b) it is composed of trivial multivalued dependencies (MVDs).

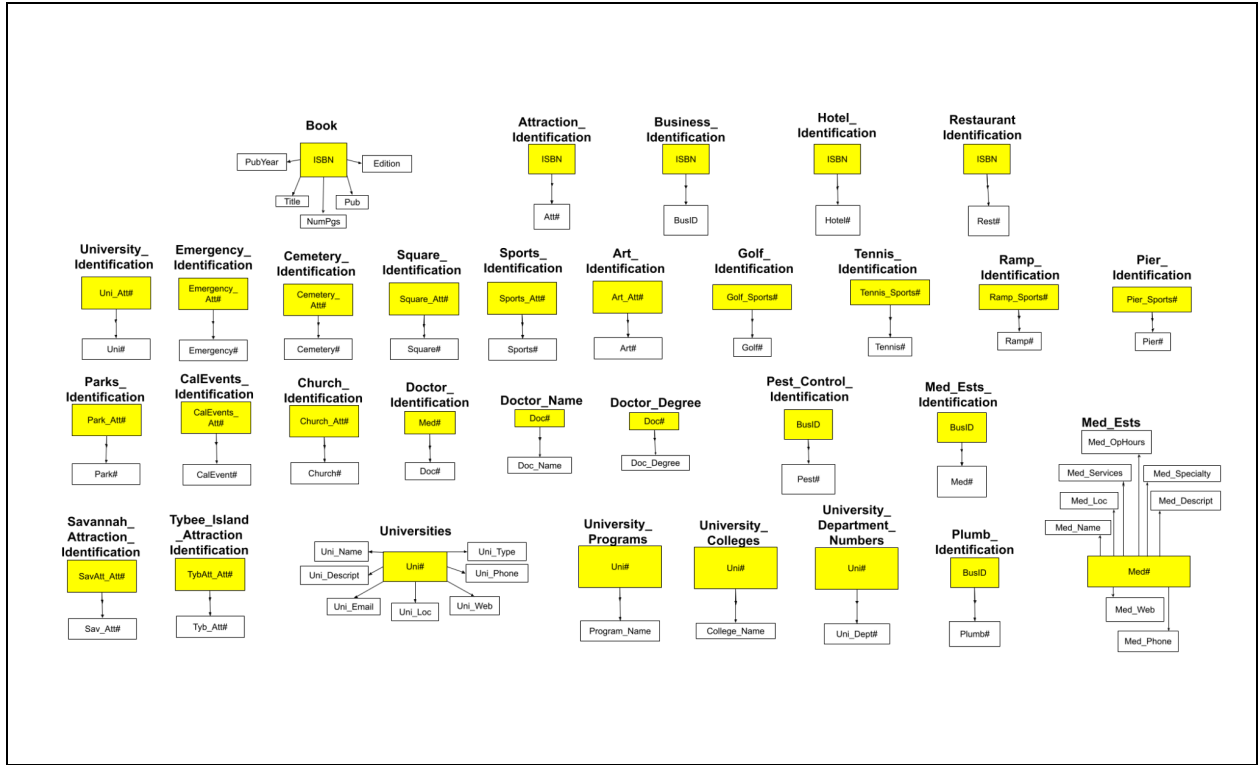


Figure 7: Diagram of the relations in 4NF

Reasonings: There were multiple relations with multivalued dependencies that were decomposed. Relation **Attractions** shares MVDs with all of its attributes, so each attributes were turned into their own relations: **Attraction_Identification**, **Business_Identification**, **Hotel_Identification**, **Restaurant_Identification**, **University_Identification**, **Emergency_Identification**, **Cemetery_Identification**, **Square_Identification**, **Sports_Identification**, **Art_Identification**, **Parks_Identifications**, **CalEvents_Identification**, **Savannah_Attraction_Identification**, **Tybee_Island_Attraction_Identification**, and **Universities**. The relation **Sports_Identification** was also made entirely of MVDs with its attributes, so these were decomposed into relations **Golf_Identification**, **Tennis_Identification**, **Ramp_Identification**, and **Pier_Identification** respectively. Relation **Universities** has MVDs with the attributes *Uni_Programs*, *Uni_College*, and *UniDept#*, so these relationships were decomposed into relations **University_Programs**, **University_Colleges**, and **University_Department_Numbers** respectively. The MVDs in the relation **Businesses** were broken down into relations **Med_Ests Identification**, **Pest_Control_Identification**, and **Plumb_Identification**, and **Med_Ests_Identification**'s MVD with attribute *Doc#* becomes relation **Doctor_Identification** (which *further* broke down into relation **Doctor Degree** due to the MVD with attribute *Doc_Degree*).

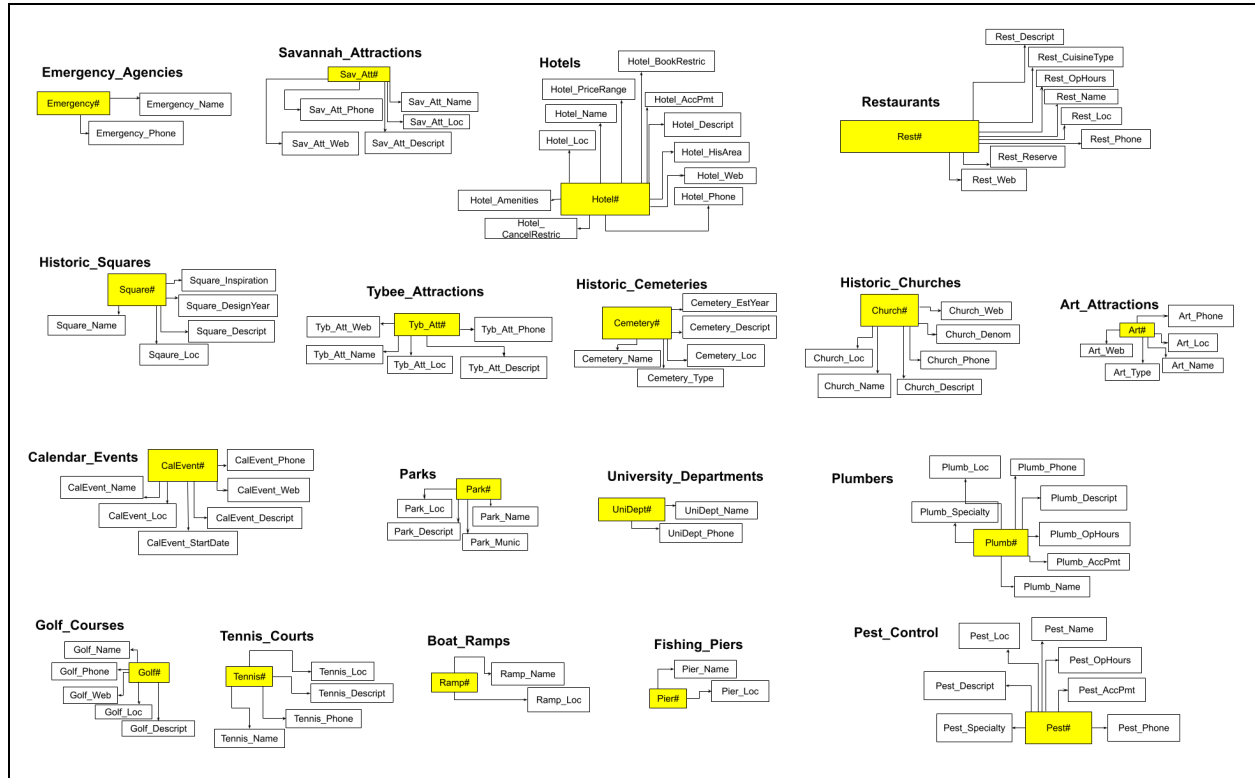


Figure 8: Diagram of remaining relations in 4NF

Final Set of Relations:

- Art_Identification(Art_Attr#, Art#)
- Art_Attractions(Art#, Art_Name, Art_Loc, Art_Phone, Art_Web, Art_Type)
- Attraction_Identification(ISBN, Attr#)
- Boat_Ramps(Ramp#, Ramp_Name, Ramp_Loc)
- Book(ISBN, Pub, Edition, PubYear, NumPgs)
- Business_Identification(ISBN, BusID)
- Calendar_Events(CalEvent#, CalEvent_Name, CalEvent_Loc, CalEvent_Phone, CalEvent_Web, CalEvent_Descript, CalEvent_StartDate)
- CalEvents_Identification(CalEvent_Attr#, CalEvents#)
- Cemetery_Identification(Cemetery_Attr#, Cemetery#)
- Church_Identification(Church_Attr#, Church#)
- Doctor_Degree(Doc#, Doc_Degree)
- Doctor_Identification(Med#, Doc#)
- Doctor_Name(Doc#, Doc_Name)
- Emergency_Agencies(Emergency#, Emergency_Name, Emergency_Phone)
- Emergency_Identification(Emergency_Attr#, Emergency#)
- Fishing_Piers(Pier#, Pier_Name, Pier_Loc)
- Golf_Courses(Golf#, Golf_Name, Golf_Phone, Golf_Web, Golf_Loc, Golf_Descript)

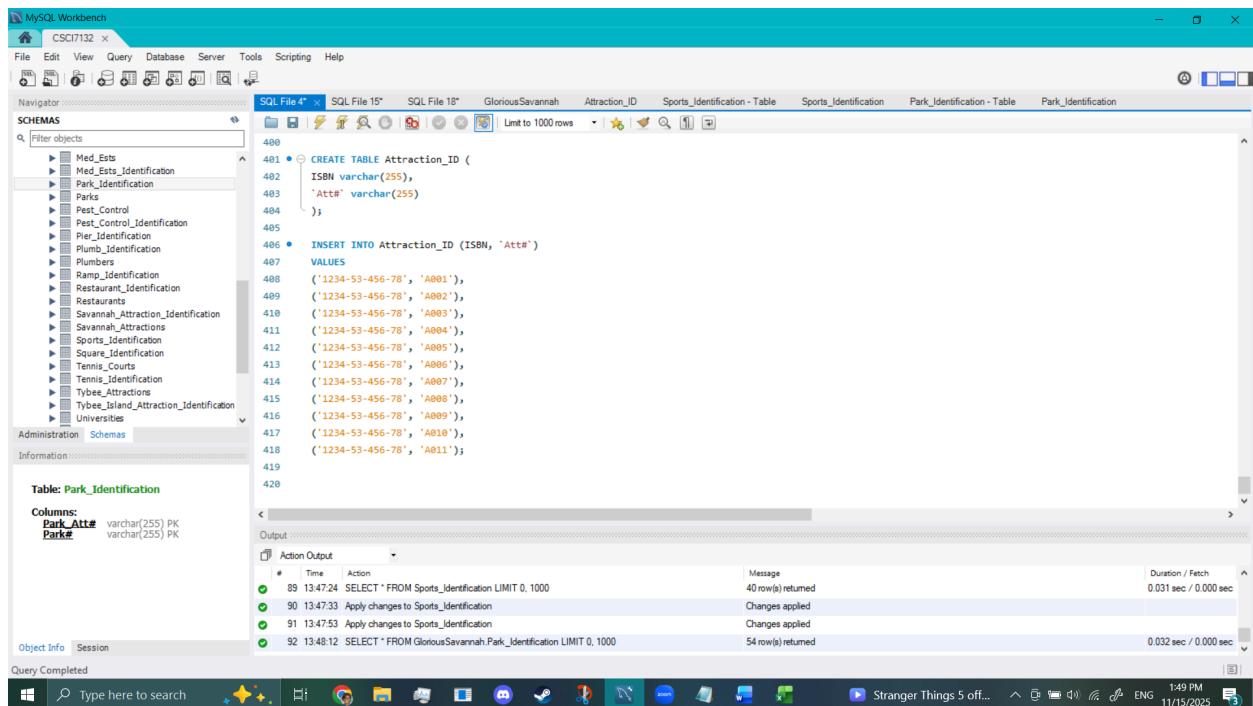
- Golf_Identification(Golf_Sports#, Golf#)
- Historic_Cemeteries(Cemetery#, Cemetery_Name, Cemetery_Loc, Cemetery_Descript, Cemetery_EstYear, Cemetery_Type)
- Historic_Churches(Church#, Church_Name, Church_Loc, Church_Descript, Church_Denom, Church_Phone, Church_Web)
- Historic_Squares(Square#, Square_Name, Square_Loc, Square_DesignYear, Square_Inspiration, Square_Descript)
- Hotel_Identification(ISBN, Hotel#)
- Hotels(Hotel#, Hotel_Name, Hotel_Loc, Hotel_Phone, Hotel_Web, Hotel_HisArea, Hotel_PriceRang, Hotel_AccPmt, Hotel_BookRestrict, Hotel_CancelRestrict, Hotel_Amenities, Hotel_Descript)
- Med_Ests(Med#, Med_Name, Med_Loc, Med_Services, Med_OpHours, Med_Specialty, Med_Web, Med_Phone, Med_Descript)
- Med_Ests_Identification(BusID, Med#)
- Park_Identification(Park_Att#, Park#)
- Parks(Park#, Park_Loc, Park_Name, Park_Descript, Park_Munic)
- Pest_Control(Pest#, Pest_Name, Pest_Loc, Pest_Phone, Pest_OpHours, Pest_AccPmt, Pest_Descript, Pest_Specialty)
- Pest_Control_Identification(BusID, Pest#)
- Pier_Identification(Pier_Sports#, Pier#)
- Plumb_Identification(BusID, Plumb#)
- Plumbers(Plumb#, Plumb_Name, Plumb_Loc, Plumb_Phone, Plumb_AccPmt, Plumb_Specialty, Plumb_OpHours, Plumb_Descript)
- Ramp_Identification(Ramp_Sports#, Ramp#)
- Restaurant_Identification(ISBN, Rest#)
- Restaurants(Rest#, Rest_Name, Rest_Loc, Rest_CuisineType, Rest_Reserve, Rest_Descript, Rest_OpHours, Rest_Web, Rest_Phone)
- Savannah_Attraction_Identification(SavAtt_Att#, Sav_Att#)
- Savannah_Attractions(Sav_Att#, Sav_Att_Name, Sav_Att_Loc, Sav_Att_Web, Sav_Att_Phone, Sav_Att_Descript)
- Sports_Identification(Sports_Att#, Sports#)
- Square_Identification(Square_Att#, Square#)
- Tennis_Courts(Tennis#, Tennis_Name, Tennis_Loc, Tennis_Phone, Tennis_Descript)
- Tennis_Identification(Tennis_Sports#, Tennis#)
- Tybee_Attractions(Tyb_Att#, Tyb_Att_Name, Tyb_Att_Loc, Tyb_Att_Phone, Tyb_Att_Web, Tyb_Att_Descript)
- Tybee_Island_Attraction_Identification(TybAtt_Att#, Tyb_Att#)
- Universities(Uni#, Uni_Name, Uni_Loc, Uni_Type, Uni_Phone, Uni_Web, Uni_Descript)
- University_Colleges(Uni#, Uni_College)

- University_Department_Numbers(Uni#, Uni_Dept#)
- University_Departments(UniDept#, UniDept_Name, UniDept_Phone)
- University_Identification(Uni_Att#, Uni#)
- University_Programs(Uni#, Uni_Program)

Implementation

Insertion

Attractions



Attraction_Identification